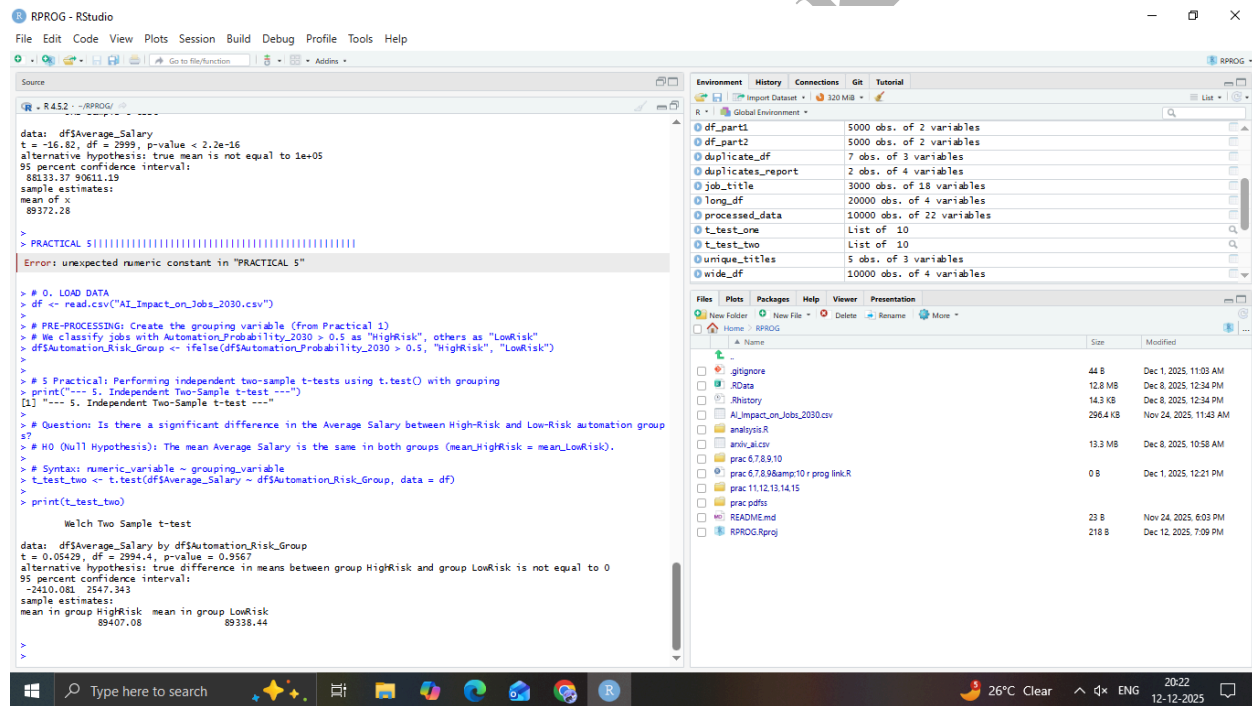


SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME : DATA ANALYSIS WITH SAS/SPSS/R

PRACTICAL 5

AIM: Performing independent two-sample t-tests using `t.test()` with grouping



```
> R452 - RPROG.R
data: df$Average_Salary
t = -16.82, df = 2999, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 1e+05
95 percent confidence interval:
 86133.37 90611.19
sample estimates:
mean of x
 89372.28

> PRACTICAL 5|||||||||||||||||||||||||||||||||||||||||
Error: unexpected numeric constant in "PRACTICAL 5"

> # 0. LOAD DATA
> df <- read.csv("AI_Impact_on_Jobs_2030.csv")
>
> # PRE-PROCESSING: Create the grouping variable (from Practical 1)
> # We classify jobs with AutomationProbability_2030 > 0.5 as "HighRisk", others as "LowRisk"
> df$Automation_Risk_Group <- ifelse(df$AutomationProbability_2030 > 0.5, "HighRisk", "LowRisk")
>
> # $ Practical: Performing independent two-sample t-tests using t.test() with grouping
> print("--- $: Independent Two-Sample t-test ---")
[1] "--- $: Independent Two-Sample t-test ---"
>
> # Question: Is there a significant difference in the Average Salary between High-Risk and Low-Risk automation group?
> # H0 (Null Hypothesis): The mean Average Salary is the same in both groups (mean_HighRisk = mean_LowRisk).
>
> # Syntax: numeric_variable ~ grouping_variable
> t_test_two <- t.test(df$Average_Salary ~ df$Automation_Risk_Group, data = df)
> print(t_test_two)

Welch Two Sample t-test

data: df$Average_Salary by df$Automation_Risk_Group
t = 0.05429, df = 2994.4, p-value = 0.9567
alternative hypothesis: true difference in means between group HighRisk and group LowRisk is not equal to 0
95 percent confidence interval:
 -2410.081 2547.343
sample estimates:
mean in group HighRisk mean in group LowRisk
 89407.08 89338.44

>
>
```

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S086